

Improving Preparedness of Communities for Evacuations using ZEVs

Definition of use cases for Merced/Mariposa evacuations

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Problem Statement

- Due to the rapid increase of ZEV ownership in recent years, rural areas like in Mariposa are facing a big challenge in the evacuation using ZEVs.
 - These rural areas normally lack charging facilities to support mass evacuations using ZEVs.
 - The transportation networks are vulnerable to support mass evacuations.
 - Very limited routes to nearby cities.
 - These routes normally consist of state highways that are 2-lane roadways. Therefore, any traffic incident happening on the roadway will significantly degrade the evacuation effectiveness and efficiency.
 - Changes to traffic control at arterial intersections are needed to support the large volume of vehicles traveling in the dominated direction.

Problem Statement (cont.)

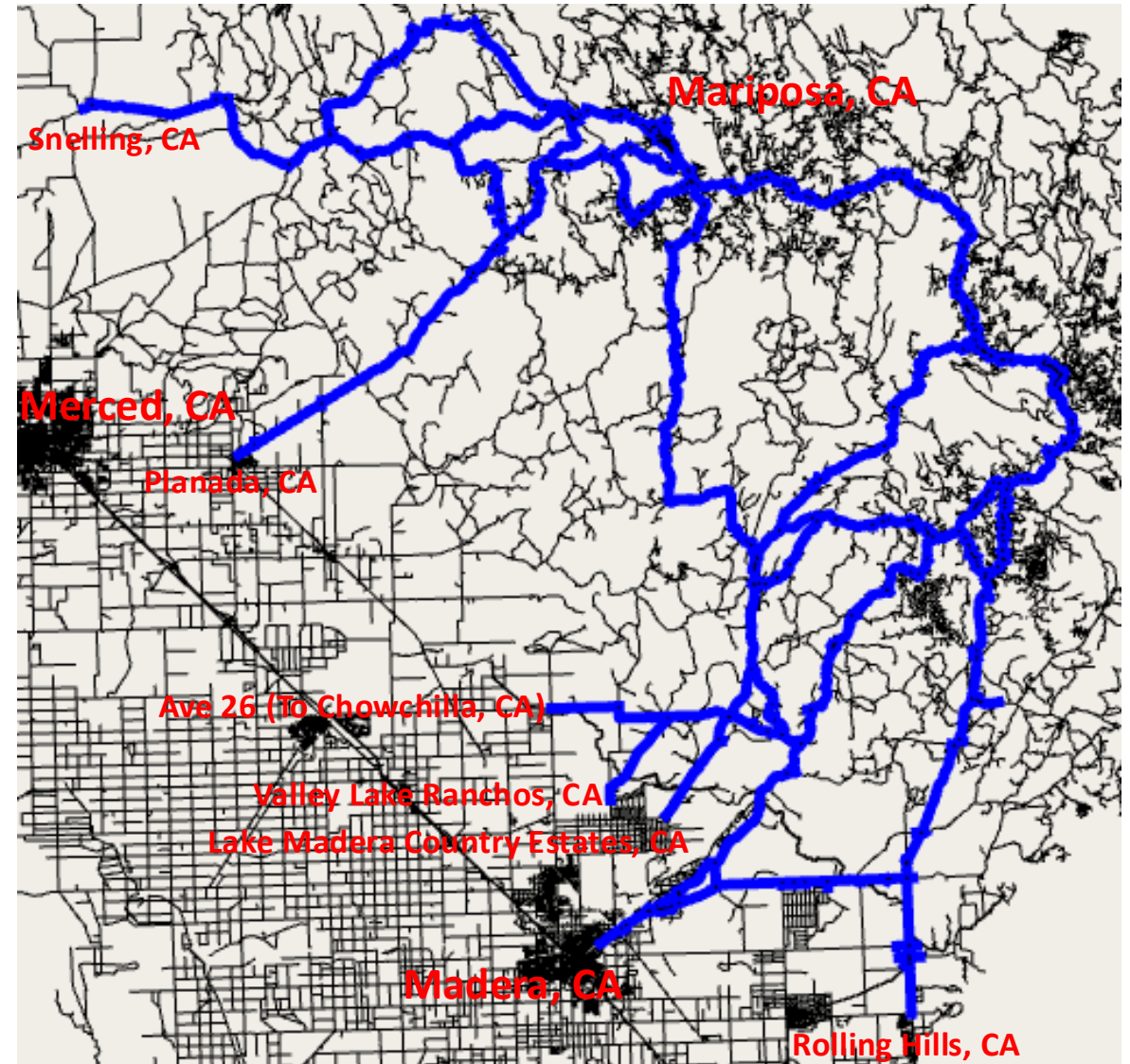
- Once the evacuation plans for ZEVs are developed (normally at the planning/macroscopic level), their performance needs to be evaluated at the operational/microscopic level.
 - The planning/macroscopic tools may not be able to appropriately capture
 - Abnormal traffic congestion at arterial intersections due to the factors like excessive left-turn demands and low intersection throughputs (e.g., at stop signs).
 - The impact of traffic incidents, e.g., reduced throughputs, longer travel delays, road blockages, gridlocks, etc.

Proposed Solution

- In this project, we want to develop a microsimulation platform in Aimsun (<https://www.aimsun.com/>) to support the evaluation of ZEV evacuation plans in the Merced/Mariposa area, which consists of the following:
 - A calibrated microsimulation model for the Merced/Mariposa network.
 - An enhanced simulation platform with novel features to support
 - The modeling and calculation of EV battery consumption in a more dynamic and realistic environment;
 - The modeling and management of fixed/mobile charging stations;
 - The routing of EVs for charging;
 - The modeling and assessment of evacuation plans (e.g., activation/deactivation of new traffic control along the evacuation routes, vehicle routing, performance measures, etc.)

Current Progress

- We have created a microsimulation network in the Merced/Mariposa area.
- While creating the network,
 - We focused on the scenario of ZEV evacuations from Mariposa to nearby cities (e.g., Merced, Madera, etc.)
 - We modeled the major arterials inside Mariposa, the state highways, and other 2-lane roadways.



Current Progress (cont.)

- We are in the process of calibrating this microsimulation model.
- We need your support if you happen to have or know how to get the following types of data for the study site:
 - Travel demand models (for Origin-Destination demands)
 - Timing sheets and traffic counts (or the Synchro model) at the signalized intersections
 - Traffic counts (or even the AADT) along the state highways and the 2-lane roadways (we might be able to get some data from the PeMS system)
 - Other data like probe travel times

Current Progress (cont.)

- Meanwhile, we have finished the architecture design for the modeling of EV charging at fixed/mobile charging stations in Aimsun.
- We are currently implementing the required data structures and workflows.
- We hope to show this enhanced simulation platform in our next workshop.

Q&A?